Week 1: Gradient-based Algorithms for Optimization

Recall that an optimization problem in standard form is given by

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0, i \in [m] := \{1, 2, \ldots, m\}, \\ & h_j(x) = 0, j \in [p]. \end{aligned}$$

Most algorithms generate a sequence $x_0, x_1, x_2, \ldots$ by exploiting local information collected on the path.

- Zeroth Order: Only $f(x_t), g_i(x_t), h_j(x_t)$ available.
- First Order: Gradients $\nabla f(x_t), \nabla g_i(x_t), \nabla h_j(x_t)$ are used. Heavily used in ML.
- Second Order: Hessian information is used. Eg: Newton's Method, etc.

ML extensively uses only first order methods

$x_0, x_1, x_2, \ldots$

Measure of progress

Let $x^\star$ be the optimal solution. The iterative algorithms continue till any of the following error metrics is sufficiently small.

- $\texttt{err}_t := ||x_t - x^\star||$ ≥ 0

  These error measures just signify when to stop the algorithm

- $\texttt{err}_t := f(x_t) - f(x^\star)$ ≥ 0

- A solution $\bar{x}$ is ϵ-optimal when

$$f(\bar{x}) \leq f(x^\star) + \epsilon.$$

  checking upon the value of function to stop the algo

  We often run the algorithm till $\texttt{err}_t$ is smaller than a sufficiently small $\epsilon > 0$.

- In presence of constaints, we define

$$\texttt{err}_t := \max(f(x_t) - f(x^\star), g_1(x_t), g_2(x_t), \ldots, g_m(x_t), |h_1(x_t)|, \ldots, |h_p(x_t)|).$$

$$\begin{bmatrix} x_1^1 \\ x_1^2 \\ \vdots \\ x_1^n \end{bmatrix} = \begin{bmatrix} x_0^1 \\ x_0^2 \\ \vdots \\ x_0^n \end{bmatrix} - \eta \begin{bmatrix} \frac{\partial f}{\partial x_1}(x_0) \\ \partial f/\partial x_2 (x_0) \\ \vdots \end{bmatrix}$$

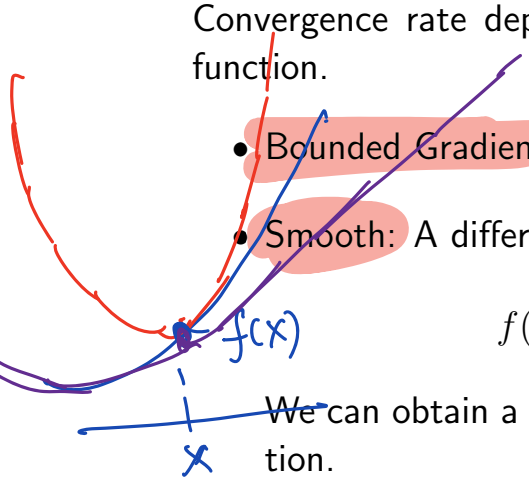
First order methods: Gradient descent

Consider the unconstrained optimization problem: $\min_{x \in \mathbb{R}^n} f(x)$

> Gradient Descent (GD): $x_{t+1} = x_t - \eta_t \nabla f(x_t), \quad t \geq 0$ starting from an initial guess $x_0 \in \mathbb{R}^n$.

The stationarity condition satisfies $x^* = x^* - \eta_t \nabla f(x^*) \implies \nabla f(x^*) = 0$.

Convergence rate depends on choice of step size η_t and characteristic of the function.



- Bounded Gradient: $||\nabla f(x)|| \leq G$ for all $x \in \mathbb{R}^n$.
- Smooth: A differentiable convex f is β-smooth if for any x, y, we have

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2}||y - x||^2. \quad \leftarrow l_x(y)$$

  We can obtain a quadratic upper bound on the function from local information.

- Strongly Convex: A differentiable convex f is α-strongly convex if for any x, y, we have

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2}||y - x||^2.$$

  We can obtain a quadratic lower bound on the function from local information.

- If f is twice differentiable, then

  both β and α need to be strictly positive.

  - f is β-smooth if and only if $\nabla^2 f(x) \preceq \beta I$ or $\lambda_{\max}(\nabla^2 f(x)) \leq \beta$ for all $x \in \mathbb{R}^n$.
  - f is α-strongly convex if and only if $\nabla^2 f(x) \succeq \alpha I$ or $\lambda_{\min}(\nabla^2 f(x)) \geq \alpha$ for all $x \in \mathbb{R}^n$.

- Determine β and α for $f(x) = ||Ax - b||_2^2 = (Ax-b)^T(Ax-b)$

$$= \underbrace{x^T A^T A x - 2b^T A x + b^T b}_{x^T Q x + r^T x}$$

$\nabla f(x) = 2A^T A x - 2A^T b.$

$\nabla^2 f(x) = 2A^T A.$: always positive semidefinite.

Gradient Descent with Bounded Gradient Assumption

skipped

Let $x_0, x_1, \ldots, x_{T-1}$ be the iterates generated by the GD algorithm.

For any t, we define $\widehat{x}_t := \frac{1}{t}\sum_{i=0}^{t-1} x_i$. Let $x^\star$ be the optimal solution.

Theorem 1: Convergence of Gradient Descent

Let the function f satisfy the $||\nabla f(x)|| \leq G$ for all $x \in \mathbb{R}^n$. Let $||x_0 - x^\star|| \leq D$. Then, for the choice of step size $\eta_t = \frac{D}{G\sqrt{T}}$, we have

$$f(\widehat{x}_T) - f(x^\star) \leq \frac{DG}{\sqrt{T}}.$$

To find an ϵ optimal solution, choose $T \geq \left(\frac{DG}{\epsilon}\right)^2$ and $\eta = \frac{\epsilon}{G^2}$.

Possible Limitation: Need to know G and D.

$x_{t+1} = x_t - \eta \nabla f(x_t)$

Gradient Descent with Smoothness Assumption

Recall that a differentiable convex f is β-smooth if for any x, y, we have

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2}||y - x||^2.$$

Theorem 2

Let the function f be β-smooth. Let $||x_0 - x^\star|| \leq D$. Then, for the choice of step size $\eta_t = \frac{1}{\beta}$, we have

$$f(x_T) - f(x^\star) \leq \frac{\beta ||x_0 - x^\star||^2}{2T}.$$

Proof: Define the following (potential) function:

$$\Phi_t := t[f(x_t) - f(x^\star)] + \frac{\beta}{2}||x_t - x^\star||^2.$$

We can show that Φ_t is decreasing in t which yields the bound.

Gradient Descent with Smoothness and Strong Convexity

Recall that a differentiable convex f is α-strongly convex if for any x, y, we have

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2}||y - x||^2.$$

Theorem 3

Let the function f be β-smooth and α-strongly convex with $\alpha \leq \beta$. Define condition number $\kappa := \frac{\beta}{\alpha}$. Then, for the choice of step size $\eta_t = \frac{1}{\beta}$, we have

$$f(x_T) - f(x^\star) \leq (1+\gamma)^{-T}(f(x_0) - f(x^\star)) \leq e^{-\frac{T}{\kappa}}(f(x_0) - f(x^\star)).$$

Note: To obtain ϵ-optimal solution, choose $T = \mathcal{O}\left(\log(\frac{1}{\epsilon})\right)$.

Proof: Define the following (potential) function:

$$\Phi_t := (1+\gamma)^t[f(x_t) - f(x^\star)], \qquad \text{where} \quad \gamma = \frac{1}{\kappa - 1} = \frac{\alpha}{\beta - \alpha}.$$

It can be shown that $\Phi_{t+1} \leq \Phi_t$.

Summary of gradient descent convergence rates

Consider the unconstrained optimization problem: $\min_{x \in \mathbb{R}^n} f(x)$

Gradient Descent (GD): $x_{t+1} = x_t - \eta_t \nabla f(x_t), \quad t \geq 0$ starting from an initial guess $x_0 \in \mathbb{R}^n$.

Theorem 4: GD Convergence rates

Let $||x_0 - x^\star|| \leq D$.

- If $||\nabla f(x)|| \leq G$ for all $x \in \mathbb{R}^n$, then with $\eta_t = \frac{D}{G\sqrt{T}}$, $f(\widehat{x}_T) - f(x^\star) \leq \frac{DG}{\sqrt{T}}$.
- If f is β-smooth, for $\eta_t = \frac{1}{\beta}$, $f(x_T) - f(x^\star) \leq \frac{\beta ||x_0 - x^\star||^2}{2T}$.
- If f is β-smooth and α-strongly convex, for $\eta_t = \frac{1}{\beta}$, $f(x_T) - f(x^\star) \leq e^{-\frac{T}{\kappa}}(f(x_0) - f(x^\star))$ where $\kappa := \frac{\beta}{\alpha}$ is the condition number.

Accelerated Gradient Descent

Start with $x_0 = y_0$. At every state t,

$$y_{t+1} = x_t - \frac{1}{\beta}\nabla f(x_t)$$
$$x_{t+1} = (1 + \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1})y_{t+1} - \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}y_t$$

Theorem 5

Let f be β-smooth, α-strongly convex with $\kappa = \frac{\beta}{\alpha}$ and let $\gamma = \frac{1}{\sqrt{\kappa}-1}$. Then, we have

$$f(y_T) - f(x^\star) \leq (1+\gamma)^{-T}\Big(\frac{\alpha+\beta}{2}||x_0 - x^*||^2\Big).$$

Improvement upon the previous rate where we had $\gamma = \frac{1}{\kappa-1}$.

Momentum Method:

$$x_{t+1} = x_t - \eta\nabla f(x_t) + \mu(x_t - x_{t-1}),$$

where $\eta = \frac{1}{\beta}$ when the function is smooth, and $\mu \in (0,1)$.